

PR 1

- `pandas` → Used for data handling and `DataFrame` operations.
- `numpy` → Used for numerical operations.
- `LabelEncoder` → Converts categorical text values into numeric values.
- `MinMaxScaler` → Normalizes numerical values between 0 and 1.

1. What is Pandas?

Pandas is a Python library used for data manipulation and analysis.

2. What is a DataFrame?

A `DataFrame` is a two-dimensional table-like data structure in Pandas.

3. Why do we use NumPy?

NumPy is used for numerical and mathematical operations.

4. What is data preprocessing?

Data preprocessing is the process of cleaning and transforming raw data before analysis or machine learning.

5. What does `read_csv()` do?

It reads a CSV file and loads it into a `DataFrame`.

Dataset Questions

6. What does `head()` function do?

Displays the first 5 rows of the dataset.

7. What is the purpose of `info()`?

Provides dataset structure, data types, and null values.

8. What does `shape` return?

Returns number of rows and columns.

9. Why check missing values?

Missing values can reduce model accuracy.

10. What does `describe()` show?

Shows statistical summary of numerical columns.

Data Type Questions

11. Why convert Age and Fare into float?

To ensure numerical calculations can be performed properly.

12. What is type casting?

Converting one data type into another.

Normalization Questions

13. What is normalization?

Scaling data into a fixed range, usually 0 to 1.

14. Why use MinMaxScaler?

To normalize numerical values.

15. What is the range of MinMaxScaler?

0 to 1.

16. Why is normalization important?

It improves machine learning performance and prevents bias toward large values.

Encoding Questions

17. What is Label Encoding?

Converting categorical text data into numeric values.

18. Why convert categorical data into numbers?

Machine learning algorithms require numeric input.

19. What is the role of LabelEncoder?

It assigns numeric labels to categorical values.

Advanced Questions

20. What happens if missing values are not handled?

The model may produce inaccurate results or errors.

21. Difference between normalization and standardization?

Normalization	Standardization
Scales between 0 and 1	Converts to mean = 0 and std = 1
Uses MinMaxScaler	Uses StandardScaler

22. What are categorical variables?

Variables containing text categories like gender, city, etc.

23. What is machine learning preprocessing?

Preparing raw data into clean formatted data suitable for machine learning algorithms.

24. What is the use of `fit_transform()`?

- `fit()` learns parameters
- `transform()` applies transformation
- `fit_transform()` does both together

25. Why is preprocessing important in Data Science?

Because clean and formatted data improves model accuracy and efficiency.

PR 2

Why Mean is Used?

Mean is commonly used because:

- It keeps data balanced
- Easy to calculate
- Suitable for numerical data

What is Z-score?

Z-score tells how far a value is from the mean.

Formula:

Why Use `np.abs()` ?

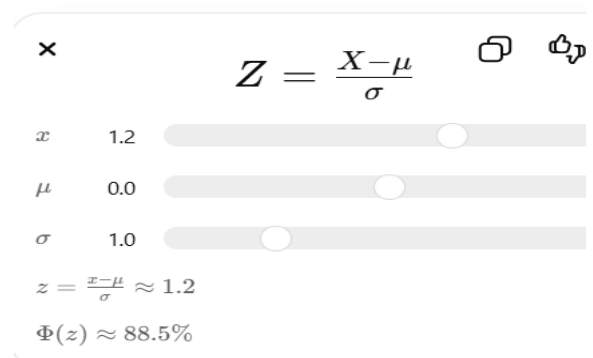
Converts negative Z-scores into positive values.

Why Normalization?

Normalization:

- Scales data uniformly
- Improves machine learning performance
- Removes magnitude differences

Why Log Transformation?



Where:

- X = data value
- μ = mean
- σ = standard deviation

Used to:

- Reduce skewness
- Handle large values
- Make data more normally distributed

Complete Workflow of the Program

Input

Student academic dataset

Processing

- Detect missing values
- Replace missing values
- Detect outliers using Z-score
- Remove outliers
- Normalize attendance
- Apply log transformation

Output

Clean dataset ready for analysis or machine learning

7. What is another method to handle missing values?

- Median
- Mode
- Delete rows/columns

Outlier Questions

8. What is an outlier?

An unusually large or small value compared to other data.

9. What is Z-score?

It measures how far a value is from the mean.

10. What is the Z-score threshold for outliers?

Usually:

$Z\text{-score} > 3$

11. Why was 400 considered an outlier?

Because student marks normally cannot exceed 100.

12. Why use median for outlier replacement?

Median is less affected by extreme values.

Normalization Questions

13. What is normalization?

Scaling values into a fixed range.

14. What is the range in MinMaxScaler?

0 to 1.

15. Why normalize attendance?

To make values comparable and improve ML performance.

Transformation Questions

16. What is log transformation?

Applying logarithm to data values.

17. Why use log transformation?

To reduce skewness and large value impact.

Library Questions

18. What is the role of SciPy?

Used for scientific and statistical calculations.

19. What does `stats.zscore()` do?

Calculates Z-score for data.

20. What does `np.where()` do?

Finds positions satisfying a condition.

Advanced Questions

21. Difference between mean and median?

Mean	Median
Average value	Middle value
Affected by outliers	Not affected much

22. Why is preprocessing important?

Because clean data improves model accuracy.

23. What is skewness?

Uneven distribution of data.

24. What happens if outliers are not removed?

They can negatively affect analysis and ML models.

25. What is the purpose of MinMaxScaler?

To scale numerical data between 0 and 1.

Mean formula

$$\bar{x} = \frac{\sum x}{n}$$

Where:

- $\bar{x}$ = mean
- x = values
- n = total values

Standard Deviation Formula

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$$

Where:

- σ = standard deviation
- μ = mean
- N = number of observations



Complete Workflow of the Program

Input

Iris dataset (`iris_3.csv`)

Processing

- Load dataset
- Display dataset information
- Group by species
- Calculate statistics
- Extract numeric values
- Generate detailed analysis

Output

Species-wise statistical summary of Iris flowers.

Important Concepts Used

Concept	Purpose
DataFrame	Store tabular data
GroupBy	Group records
Aggregation	Apply statistical functions
Mean	Average value
Median	Middle value
Standard Deviation	Measure spread
Filtering	Select rows based on condition

Viva / Oral Questions with Answers

Basic Questions

1. What is the Iris dataset?

The Iris dataset contains measurements of iris flowers used in machine learning.

What does `groupby()` do?

Groups data based on a common column.

9. Why group by species?

To analyze each flower species separately.

10. What is aggregation?

Applying statistical functions on grouped data.

Statistical Questions

11. What is mean?

Average of values.

12. What is median?

Middle value after sorting data.

13. What is standard deviation?

Measures how spread out values are.

14. What is minimum value?

Smallest value in a dataset.

15. What is maximum value?

Largest value in a dataset.

Data Analysis Questions

16. What does `describe()` do?

Displays statistical summary of numerical columns.

17. What is filtering in Pandas?

Selecting rows based on conditions.

Example:

```
df[df['species'] == 'setosa']
```

18. What does `unique()` do?

Returns distinct values from a column.

19. What does `apply(list)` do?

Converts grouped values into lists.

Advanced Questions

20. Why is statistical analysis important?

It helps understand patterns and distribution of data.

21. What is exploratory data analysis (EDA)?

Initial analysis performed to understand data.

22. Difference between mean and median?

Mean	Median
Average value	Middle value
Affected by outliers	Less affected

23. What is variance?

Measure of data spread.

24. Why use grouping in data analysis?

To compare categories separately.

25. What are the three species in Iris dataset?

- Setosa
- Versicolor
- Virginica

PR 4

train_test_split

Splits dataset into training and testing data.

LinearRegression

Machine learning algorithm used for prediction.

Metrics

Used to evaluate model performance.

Why Split Data?

- Training data teaches the model
- Testing data checks model performance

```
random_state=42
```

Ensures same random split every time.

Mean Absolute Error (MAE)

Python

```
mae = mean_absolute_error(y_test, y_pred)
```

Formula

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

Meaning

Average absolute prediction error.

Lower MAE = better model.

Mean Squared Error (MSE)

Python

R² Score

Python

```
r2 = r2_score(y_test, y_pred)
```

Formula

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

Meaning

Measures how well model explains variance.

R² Value

Meaning

1

Perfect prediction

0

Poor prediction

+ Ask anything



Train-Test Split Model validation

MAE Average error

MSE Squared error

RMSE Root error

R² Score Accuracy measure

Root Mean Squared Error (RMSE)

”

Python

```
rmse = np.sqrt(mse)
```

Formula

$$RMSE = \sqrt{MSE}$$

Meaning

Measures prediction accuracy in original units.

What is Linear Regression?

A supervised machine learning algorithm used to predict continuous values.

2. What is the target variable in this program?

MEDV – **Median house value.**

3. What are independent variables?

Input features used for prediction.

4. What is supervised learning?

Learning using labeled input-output data.

5. Why use Linear Regression?

To predict numerical values.

Dataset Questions

6. What does `read_csv()` do?

Loads CSV file into DataFrame.

7. Why check missing values?

Missing values affect model performance.

8. What does `fillna()` do?

Replaces missing values.

Splitting Questions

9. Why split dataset?

To test model performance on unseen data.

10. What is training data?

Data used to train model.

11. What is testing data?

Data used to evaluate model.

Model Questions

12. What does `fit()` do?

Trains the model.

13. What does `predict()` do?

Generates predictions.

14. What is coefficient in Linear Regression?

Represents impact of feature on target.

15. What is intercept?

Prediction value when all inputs are zero.

Evaluation Questions

16. What is MAE?

Average absolute error.

17. What is MSE?

Average squared error.

18. What is RMSE?

Square root of MSE.

19. What is R^2 score?

Measures model accuracy.

20. Which is better: high or low RMSE?

Low RMSE is better.

Visualization Questions

21. Why use scatter plot?

To compare actual vs predicted values.

22. What indicates good prediction in scatter plot?

Points close to straight diagonal line.

Advanced Questions

23. Difference between classification and regression?

Classification	Regression
-----------------------	-------------------

Predict categories	Predict continuous values
--------------------	---------------------------

24. What is overfitting?

Model performs well on training data but poorly on test data.

25. What is machine learning model evaluation?

Process of measuring prediction performance.

PR 5

Term	Meaning
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TP	Correctly predicted purchase
----	------------------------------

TN	Correctly predicted no purchase
----	---------------------------------

FP	Wrongly predicted purchase
----	----------------------------

FN	Missed actual purchase
----	------------------------

Library	Purpose
----------------	----------------

train_test_split	Split dataset
------------------	---------------

StandardScaler	Feature scaling
----------------	-----------------

Library	Purpose
LogisticRegression	Classification model
confusion_matrix	Performance evaluation
accuracy_score	Calculate accuracy
precision_score	Calculate precision
recall_score	Calculate recall

Why Feature Scaling?

Feature scaling:

- Makes all variables comparable
- Improves model performance
- Speeds up learning

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

Meaning

Percentage of correct predictions.

Error Rate

```
<> Python
error_rate = 1 - accuracy
```

Formula:

$$Error\ Rate = 1 - Accuracy$$

$$Precision = \frac{TP}{TP+FP}$$

Meaning

How many predicted positives are actual

Recall

```
<> Python
recall = recall_score(y_test, y_pred)
```

Formula:

$$Recall = \frac{TP}{TP+FN}$$

Concept	Purpose
Logistic Regression	Classification

Concept	Purpose
Feature Scaling	Standardize features
Sigmoid Function	Probability prediction
Confusion Matrix	Classification evaluation
Accuracy	Correct prediction rate
Precision	Positive prediction quality
Recall	Positive detection rate

What is Logistic Regression?

A supervised machine learning algorithm used for classification problems.

2. Is Logistic Regression used for classification or regression?

Classification.

3. What is the target variable in this program?

Purchased

4. What are independent variables?

Age and Estimated Salary.

5. What is supervised learning?

Learning using labeled input-output data.

Dataset Questions

6. What does `read_csv()` do?

Reads CSV file into DataFrame.

7. Why split dataset?

To evaluate model performance on unseen data.

8. What is testing data?

Data used to test model accuracy.

Feature Scaling Questions

9. Why use StandardScaler?

To standardize features.

10. What is feature scaling?

Making variables comparable by transforming values.

11. What happens without scaling?

Model performance may decrease.

Logistic Regression Questions

12. What is sigmoid function?

Function that converts values into probabilities between 0 and 1.

13. What is probability in Logistic Regression?

Chance of belonging to a class.

14. What does `fit()` do?

Trains the model.

15. What does `predict()` do?

Predicts output classes.

Confusion Matrix Questions

16. What is confusion matrix?

Table used to evaluate classification performance.

17. What is True Positive?

Correct positive prediction.

18. What is False Positive?

Incorrect positive prediction.

19. What is False Negative?

Incorrect negative prediction.

20. What is True Negative?

Correct negative prediction.

Metric Questions

21. What is accuracy?

Percentage of correct predictions.

22. What is precision?

Correctness of positive predictions.

23. What is recall?

Ability to detect actual positives.

24. What is error rate?

Percentage of wrong predictions.

Advanced Questions

25. Difference between Linear Regression and Logistic Regression?

Linear Regression	Logistic Regression
--------------------------	----------------------------

Predicts continuous values	Predicts categories
----------------------------	---------------------

Uses straight line	Uses sigmoid curve
--------------------	--------------------

PR 6

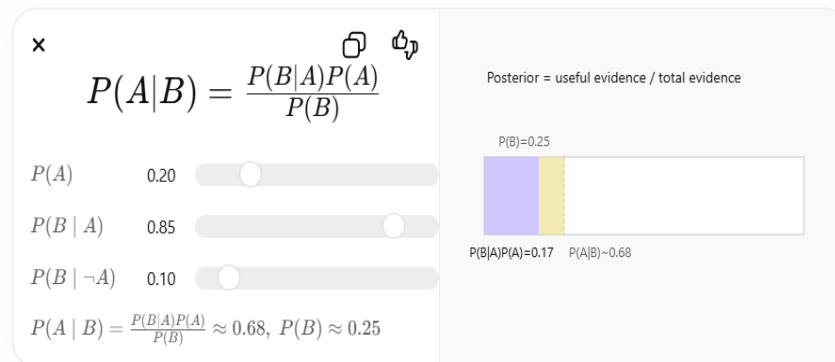
Library	Purpose
train_test_split	Split dataset
GaussianNB	Naive Bayes classifier
confusion_matrix	Evaluate classification
accuracy_score	Measure accuracy
precision_score	Measure precision
recall_score	Measure recall

What is Naive Bayes?

Naive Bayes is a:

- Supervised machine learning algorithm
- Based on Bayes Theorem
- Used for classification

Bayes Theorem



Where:

- $P(A|B)$ = Probability of A given B
- $P(B|A)$ = Probability of B given A
- $P(A)$ = Prior probability
- $P(B)$ = Evidence probability



. What is Naive Bayes?

A supervised machine learning classification algorithm based on Bayes

theorem.

2. Why is it called “Naive”?

Because it assumes all features are independent.

3. What type of algorithm is Naive Bayes?

Classification algorithm.

4. What dataset is used here?

Iris dataset.

5. What are the target classes?

- Setosa
 - Versicolor
 - Virginica
-

Dataset Questions

6. What are features in this program?

Sepal and petal measurements.

7. What is target variable?

Species column.

8. What does `iloc` do?

Selects rows and columns by index position.

Splitting Questions

9. Why split dataset?

To evaluate model performance.

10. What is training data?

Data used to train model.

11. What is testing data?

Data used to test model.

Naive Bayes Questions

12. What is Bayes theorem?

Probability theorem used for prediction.

13. What is GaussianNB?

Naive Bayes model for continuous numerical data.

14. Why use GaussianNB for Iris dataset?

Because Iris features are continuous values.

15. What assumption does Naive Bayes make?

Features are independent.

Confusion Matrix Questions

16. What is confusion matrix?

Table used to evaluate classification performance.

17. What is True Positive?

Correct positive prediction.

18. What is False Positive?

Wrong positive prediction.

19. What is False Negative?

Wrong negative prediction.

20. What is True Negative?

Correct negative prediction.

Metric Questions

21. What is accuracy?

Percentage of correct predictions.

22. What is precision?

Correctness of positive predictions.

23. What is recall?

Ability to identify actual positives.

24. What is error rate?

Percentage of wrong predictions.

Advanced Questions

25. Difference between Logistic Regression and Naive Bayes?

Logistic Regression	Naive Bayes
Discriminative model	Probabilistic model
Uses decision boundary	Uses Bayes theorem
Needs more training data	Works well on small datasets

PR 7

1. What is Text Analytics?

Text analytics is the process of extracting meaningful information from unstructured text data.

2. What is NLP?

NLP (Natural Language Processing) is a branch of AI that helps computers understand human language.

3. What is Tokenization?

Tokenization is the process of splitting text into words or sentences.

4. What are Tokens?

Individual words or units obtained after tokenization are called tokens.

5. What is POS Tagging?

POS tagging identifies grammatical categories such as noun, verb, adjective, etc.

6. What are Stop Words?

Commonly used words like "is", "the", "of" that usually do not add much meaning.

7. Why are Stop Words Removed?

To reduce unnecessary data and improve processing efficiency.

8. What is Stemming?

Stemming reduces words to their root form by removing suffixes.

9. What is Lemmatization?

Lemmatization converts words into meaningful base dictionary forms.

10. Difference Between Stemming and Lemmatization?

- Stemming may produce incorrect words.
- Lemmatization produces meaningful words.

11. What is Term Frequency?

Term Frequency counts how many times a word appears in a document.

12. What is TF-IDF?

TF-IDF measures the importance of a word in a document.

13. Why is TF-IDF Used?

To identify important words and reduce the weight of common words.

14. What is CountVectorizer?

CountVectorizer converts text into a matrix of word counts.

15. What is TfidfVectorizer?

TfidfVectorizer converts text into TF-IDF numerical representation.

16. Which library is used for NLP in this program?

The NLTK library is used.

17. Which library is used for machine learning text features?

Scikit-learn is used.

18. Why is Pandas used?

Pandas is used to display matrices in table format.

19. What is Unstructured Data?

Data without a predefined format such as text, emails, reviews, etc.

20. Applications of Text Analytics

- Sentiment Analysis
- Spam Detection
- Chatbots
- Search Engines
- Recommendation Systems
- Document Classification

PR 8

What is Data Visualization?

Data visualization is the graphical representation of data using charts and plots.

2. Which libraries are used in this program?

- Pandas
 - Seaborn
 - Matplotlib
-

3. What is the purpose of Pandas?

Pandas is used for data manipulation and analysis.

4. What is Seaborn?

Seaborn is a Python library for statistical data visualization.

5. What is Matplotlib?

Matplotlib is a plotting library used to create graphs and charts.

6. What is the Titanic dataset?

A famous dataset containing passenger information from the Titanic ship disaster.

7. What does `head()` do?

Displays the first 5 rows of the dataset.

8. What does `info()` do?

Displays dataset structure, data types, and non-null values.

9. What is a Histogram?

A histogram is a graph used to show frequency distribution of numerical data.

10. What is the purpose of `bins` in histogram?

Bins define the number of bars/groups in histogram.

11. What is KDE?

KDE (Kernel Density Estimation) is a smooth curve representing data distribution.

12. Why is `figsize` used?

To control graph size.

13. What does `x= 'fare '` mean?

It selects the fare column for visualization.

14. What is the use of `plt.show()` ?

Displays the graph.

15. What type of data is fare?

Numerical continuous data.

16. Why is histogram useful?

It helps understand:

- Distribution
 - Frequency
 - Skewness
 - Outliers
-

17. What is skewness in data?

Skewness shows whether data is symmetric or tilted toward one side.

18. What kind of skewness is seen in Titanic fare data?

Positive skewness.

19. What is the purpose of visualization?

To understand patterns and trends in data easily.

20. Applications of Data Visualization

- Business Analytics
- Machine Learning
- Finance
- Healthcare

- Research
- Data Science

PR 9

What is Data Visualization?

Data visualization is the graphical representation of data.

2. Which libraries are used in this program?

- Seaborn
 - Matplotlib
-

3. What is a Box Plot?

A box plot is a graph used to show statistical distribution of data.

4. What are the main parts of a box plot?

- Minimum
 - Q1
 - Median
 - Q3
 - Maximum
 - Outliers
-

5. What is Median?

Median is the middle value of ordered data.

6. What are Quartiles?

Quartiles divide data into four equal parts.

7. What is an Outlier?

An unusually high or low value compared to rest of data.

8. What is the purpose of hue in seaborn?

hue adds color grouping based on categories.

9. What does `x= 'sex'` represent?

Gender categories on X-axis.

10. What does `y= 'age'` represent?

Passenger age values on Y-axis.

11. What does `survived` indicate?

Whether passenger survived or not.

Value	Meaning
0	Not survived
1	Survived

12. Why is box plot useful?

It helps identify:

- Distribution
 - Spread
 - Median
 - Outliers
-

13. What is the use of `figsize`?

Controls graph size.

14. What does `plt.show()` do?

Displays the graph.

15. What type of data is age?

Continuous numerical data.

16. What kind of data is sex?

Categorical data.

17. What is categorical data?

Data divided into categories such as male/female.

18. What are the advantages of box plot?

- Easy comparison
- Detects outliers
- Shows spread of data
- Displays median clearly

19. Difference between histogram and box plot?

Histogram	Box Plot
Shows frequency	Shows statistical summary
Uses bars	Uses boxes
Displays distribution shape	Displays quartiles & outliers

20. Applications of Box Plot

- Data Analysis
- Machine Learning
- Statistics
- Healthcare
- Finance
- Research

PR 10

. What is Exploratory Data Analysis (EDA)?

EDA is the process of analyzing and visualizing datasets to understand patterns and relationships.

2. Which libraries are used in this program?

- Pandas
- Seaborn
- Matplotlib

3. What is the Iris dataset?

A famous dataset containing flower measurements of Iris species.

4. What are features in a dataset?

Features are columns/attributes describing data.

5. What is `head()` used for?

Displays first 5 rows of dataset.

6. What is `info()` used for?

Displays dataset structure and data types.

7. What is a Histogram?

A graph showing frequency distribution of numerical data.

8. What is a Boxplot?

A statistical graph showing distribution and outliers.

9. What are Outliers?

Extreme unusual values in data.

10. What is the purpose of `describe()`?

Displays statistical summary of dataset.

11. What is Mean?

Mean is the average value.

$$\text{Mean} = \frac{\sum x}{n}$$

12. What is Standard Deviation?

It measures spread of data around mean.

13. What is Skewness?

Skewness shows whether data is symmetric or tilted.

14. What is Numerical Data?

Data represented using numbers.

Example:

- Age
- Height

- Weight
-

15. What is Categorical Data?

Data represented in categories.

Example:

- Species
 - Gender
-

16. What is `select_dtypes()` used for?

Selects columns based on data types.

17. What is the purpose of `subplot()`?

Creates multiple plots in a single figure.

18. Why is `tight_layout()` used?

Prevents overlap between plots.

19. Why are visualizations important?

They help understand patterns and trends easily.

20. Applications of EDA

- Machine Learning
- Data Science

- Business Analytics
- Research
- Healthcare
- Finance

GROUP B

Pract 11

What is the purpose of this program?

To count frequency of words in a sentence.

2. Which programming language is used?

Java

3. What is MapReduce?

MapReduce is a programming model for processing large datasets.

4. What are the two phases of MapReduce?

1. Map Phase
 2. Reduce Phase
-

5. What is the Map Phase in this program?

Splitting words and counting occurrences.

6. What is the Reduce Phase in this program?

Displaying final word counts.

7. What is HashMap?

A data structure that stores key-value pairs.

8. Why is HashMap used?

To store words and their frequencies.

9. What does `split("\\s+")` do?

Splits string using spaces.

10. Why is `toLowerCase()` used?

To make word comparison case-insensitive.

11. What is `getOrDefault()`?

Returns value if key exists, otherwise returns default value.

12. What is the key in HashMap?

The word itself.

13. What is the value in HashMap?

Frequency/count of the word.

14. What is a frequency count?

Number of times a word appears.

15. What is Map.Entry?

Used to access key-value pairs in HashMap.

16. What is the output of this program?

Word frequencies.

17. Why is word count important in Big Data?

Used in:

- Search engines
 - Text analytics
 - NLP
 - Data mining
-

18. What is Big Data?

Large and complex data that cannot be processed using traditional methods.

19. Applications of Word Count Program

- Search Engines
 - Sentiment Analysis
 - Document Analysis
 - NLP
 - Text Mining
-

20. Advantages of MapReduce

- Fast processing
- Parallel execution
- Handles large data
- Scalable
- Efficient for Big Data

PR 12

What is the purpose of this program?

To count occurrences of log levels using MapReduce.

2. What is MapReduce?

A programming model used for processing large datasets.

3. What are the phases of MapReduce?

1. Map
 2. Shuffle and Sort
 3. Reduce
-

4. What is the Map phase?

Converts input data into key-value pairs.

5. What is the Reduce phase?

Aggregates values for each key.

6. What is log processing?

Analyzing system-generated logs.

7. What are log levels?

Categories of system messages.

Examples:

- INFO
 - ERROR
 - WARN
-

8. What does the Mapper class do?

Extracts log levels and creates key-value pairs.

9. What does the Reducer class do?

Adds frequencies of each log level.

10. What is a Key-Value Pair?

A pair consisting of:

- Key
- Associated value

Example:

`(INFO, 1)`

11. Why is `split(" ")` used?

To separate words in log lines.

12. Why is `toUpperCase()` used?

To standardize log levels.

13. What is `Collectors.groupingBy()` used for?

Groups values having same key.

14. What is Java Stream API?

A feature for processing collections efficiently.

15. What does `mapToInt().sum()` do?

Converts values to integers and calculates total sum.

16. What is the purpose of helper class?

To store key-value data.

17. What data structure is used for grouping?

Map collection.

18. Applications of MapReduce?

- Hadoop
 - Big Data
 - Search Engines
 - Analytics
-

19. What is Big Data?

Extremely large and complex data requiring advanced processing.

20. Advantages of MapReduce

- Parallel processing
- Scalability
- Efficient large-data handling
- Faster processing
- Fault tolerance

PR 13

What is the purpose of this program?

To analyze weather data and calculate averages.

2. Which programming language is used?

Java

3. What type of data format is used?

CSV (Comma-Separated Values) format.

4. What is `BufferedReader`?

A class used for efficient reading of text data.

5. What is `StringReader`?

A class used to read characters from a string.

6. Why is `split(",")` used?

To separate CSV values using commas.

7. What does `parseDouble()` do?

Converts string into double data type.

8. Why is `count++` used?

To count number of records.

9. Why is header skipped?

Because it contains column names, not actual data.

10. What is a loop used for in this program?

To process all weather records.

11. What is CSV?

A file format where values are separated by commas.

12. What is average?

Average is the sum of values divided by total count.

$$\text{Average} = \frac{\sum x}{n}$$

13. What data type is used for temperature?

double

14. Why is double used?

Because weather values contain decimal numbers.

15. What is the purpose of while loop?

To read data until end of file.

16. What is data parsing?

Converting one data format into another.

17. What is the output of this program?

Average weather statistics.

18. Applications of weather data analysis?

- Forecasting
 - Climate monitoring
 - Environmental research
 - Agriculture
-

19. What is data analytics?

Process of extracting useful information from data.

20. Advantages of this program

- Simple data processing
- Fast analysis
- Easy CSV handling
- Useful for real-world datasets